

Business Problem Statement

Walmart operates across multiple stores and locations, generating large volumes of weekly sales data. Analysing this data can help the business understand how sales performance changes across different stores, time periods, and holiday seasons. However, raw sales data alone does not provide a clear view of the patterns and factors that may influence sales performance.

The primary business problem addressed in this project is to understand Walmart's historical sales performance and identify meaningful patterns that can support better business planning and decision-making. The analysis focuses on store-level performance, yearly and monthly sales trends, holiday versus non-holiday sales, and the relationship between weekly sales and external factors such as temperature, fuel prices, CPI, and unemployment.

The analysis aims to answer key business questions such as which stores generate the highest sales, which periods show stronger sales performance, whether holiday weeks experience higher average sales, and how external economic and environmental factors relate to sales. These insights can help businesses identify sales patterns, understand store performance, plan inventory and staffing requirements, and evaluate opportunities for improving sales strategies.

To **address these questions**, the project follows a complete data analytics workflow. The dataset is cleaned and prepared using Python, analysed using SQL, and presented through an interactive Power BI dashboard. The main objective of this project is to analyse Walmart's historical weekly sales performance across **2010, 2011, and 2012** and identify key trends and patterns in the business during these three years. The analysis focuses on understanding how overall sales changed year over year, which months and stores performed better, and whether holiday periods had an impact on weekly sales.

Overall, the project demonstrates how historical retail data can be transformed into actionable business insights that support data-driven decision-making and improve understanding of retail sales performance.